Subject Description Form

Subject Code	ITC6000G
Subject Title	Textile Materials Science
Credit Value	3
Level	6
Pre-requisite / Co-requisite / Exclusion	Nil
Objectives	Provide detailed, foundational knowledge and guidance for research students to carry out in-depth study of governing principles in the structure and properties of textile materials and composites, including fiber, yarn, and fabric compositions.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Comprehend the governing principles in the structure and properties of textile materials and composites b. Apply the properties of textile material structures based on these governing principles to their research and development for new products, functionalities and applications. c. Communicate effectively in the textile materials science principles and their applications.
Subject Synopsis/ Indicative Syllabus	a. Relationship between structure and properties in fibers, yarns, fabrics and their composites b. Structures, properties, functionalities and their applications of textile materials and composites in mechanical, optical, thermal, electrical, biological and aesthetic areas
Teaching/Learning Methodology	This is a non- guided study course. The subject lecturer will deliver materials in class rooms and tutorials. The students are given considerable time for their explorations and are requested to take examination, make presentation and prepare reports based on the reading materials.
Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					
		a	b	c			
1. Coursework	40%	✓	✓	✓			
2. Examination	60%	✓	✓	✓			
Total	100%						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Coursework is used to train and assess students’ knowledge and skills learned in this subject. Examination is designed to assess the overall learning outcomes.

	The materials submitted for all assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.	
Student Study Effort Expected	Class contact:	
	▪ Lectures	15 Hrs.
	▪ Tutorials	15 Hrs.
	▪ Seminars	9 Hrs
	Other student study effort:	
	▪ Reading	45 Hrs.
	▪ Preparation of reports	36 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	1. SD Guler, M Garnno. Crafting Wearable: Blending Technology with Fashion (2016), Springer 2. X.M. Tao. Handbook of Smart Textiles (2015), Springer. 3. D Tilak. Electronic Textiles: Smart Fabrics and Wearable Technology (2015), Woodhead. 4. David J Taylor. Carr and Latham’s Technology of Clothing Manufacture (2008), Blackwell Publishing, 4th Edition. 5. Susan M. Watkins and Lucy E. Dunne. Functional clothing design: from sportswear to spacesuits (2015), New York, Bloomsbury Publishing Inc. 6. Steven G. Hayes; Praburaj Venkatraman. Materials and technology for sportswear and performance apparel (2016), Boca Raton, FL, CRC Press. 7. Edward Sazonov; Michael R. Neuman. Wearable Sensors: Fundamentals, Implementation and Applications (2015), Academic Press, Elsevier. 8. Karen L. LaBat; Karen S. Ryan. Human Body: A Wearable Product Designer's Guide (2019), Boca Raton, FL, CRC Press. 9. Dominique Paret; Pierre Crégo. Wearables, Smart Textiles and Smart Apparel (2019), ISTE Press - Elsevier. 10. Philippe Boisse. Composite Reinforcements for Optimum Performance (2020), Woodhead Publishing. 11. Andrea Ehrmann, Tuan Anh Nguyen and Phuong Nguyen Tri. Nanosensors and Nanodevices for Smart Multifunctional Textiles (2020), Elsevier. 12. Charis M. Galanakis. Biobased Products and Industries (2020), Elsevier. 13. Josep Ignasi de Llorens. Fabric Structures in Architecture (2015), Woodhead Publishing. 14. Goeran Pohl. Textiles, Polymers and Composites for Buildings (2010), Woodhead Publishing.	

	15. Mohd Shabbir, Shakeel Ahmed, Javed N. Sheikh. Frontiers of Textile Materials: Polymers, Nanomaterials, Enzymes, and Advanced Modification Techniques (2020), Scrivener Publishing LLC - Wiley Publishing.